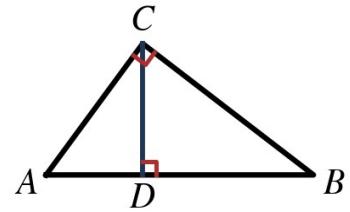


Speak Like A Mathematician:

geometric mean:

altitude to the hypotenuse:



Similar Right Triangle Theorems

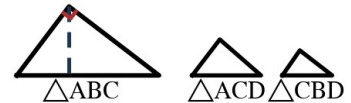
The altitude to the hypotenuse creates three _____ triangles: $\triangle ABC \sim \triangle ACD \sim \triangle CBD$ (each shares an acute angle with the original).

Geometric Mean (Altitude): $\text{altitude}^2 = (\text{segment 1}) \cdot (\text{segment 2})$ — the altitude is the geometric mean of the two hypotenuse pieces.

Geometric Mean (Leg): $\text{leg}^2 = (\text{whole hypotenuse}) \cdot (\text{segment})$ — each leg is the geometric mean of the hypotenuse and the piece TOUCHING that leg.

Example 1 - Compute Geometric Means

- Find the geometric mean of 4 and 9.
- Find the geometric mean of 5 and 20.
- Find the geometric mean of 8 and 18.



Example 2 - Name the Similar Triangles

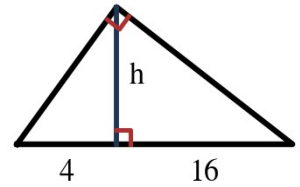
In the figure on page 1, $\overline{CD}$ is the altitude to the hypotenuse of right $\triangle ABC$.

- Write the three-triangle similarity statement.
- Why are they similar? Name the shortcut.

Example 3 - The Altitude Rule

In the figure at the right, the altitude splits the hypotenuse into segments 4 and 16.

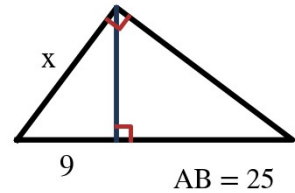
- Find h .
- Find both legs in simplest radical form (hypotenuse = 20).
- Check the legs with the Pythagorean Theorem.



Example 4 - The Leg Rule

In the figure at the right, $AB = 25$ and the segment adjacent to leg x measures 9.

- a) Find x .
- b) Find the other segment, the other leg, and the altitude.
- c) Verify with 15-20-25.



Example 5 - Solve for a Segment

In the figure at the right, the altitude is 6 and one hypotenuse segment is 4.

- a) Find x .
- b) Find the whole hypotenuse.

